Quinn Lue

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Education

Montana State University 3.76 GPA

BS, Computer Science, Minor in Math

Independent Research

Ongoing: Language Mixture Optimization for Multilingual ASR

- A study on language mixture for continual pretraining and supervised fine-tuning to optimize ASR for data-starved languages, analyzing cross-lingual transfer, linguistic grouping, data repetition scaling

Data mixture study on Audio SSL

- RegMix-style scaling study testing whether mixture quality measured at reduced scale/horizon predicts full-scale downstream performance for audio SSL, ablating budget, scale, probe vs LoRA, in vs out of distribution perf,
- With a spearman's rank of 0.698 we can say that we can compare two dataset mixtures utilizing $\approx 0.1\%$ of total FLOPs
- Identified a null correlation of 0.08 between SSL downstream performance and effective sample size up to 40x repetition, (> 0.9 for supervised pretraining)
- Calculated pretraining and readout seed variance and signal-to-noise ratio of each cell to provide statistical grounding and disattentuate correlations

Audio SSL Training Library

- A research focused audio SSL modelling library for entirely bit-reproducible training runs from a single config
- Supports clean ablations across processor, backbone, pretraining objective, augmentation policy

Projects

Deep Learning Library and LLM Pre/Post-training | inference link

- Built a deep learning library and LLM training framework on NumPy/CuPy supporting core tensor ops, checkpointing, reverse mode autograd, mixed precision, KV-cache, output streaming, LoRA, etc
- Pretrained and finetuned a small LLM (0.3b parameters, 40 A40 hrs, 8.8b tokens, 18.17 PPL on OWT2) on this library

Scrabble AI

- Achieved superhuman Scrabble performance via counter-factual regret minimization and Monte-Carlo tree search

Experience

Terminal Bench 3.0 Contributions

- Authored and reviewed machine learning tasks for Terminal Bench 3.0

Undergraduate Research Assistant

- Developed a LLM assisted pipeline for improved Risk and Crisis Communication (RCC)
- Under review Military Cyber Affairs: Persuasion with more Precision - Second Author

Technical Skills

Languages: Python
Machine Learning: PyTorch, Deep learning, Natural Language Processing (NLP), RAG, representation learning, Transformers, Generative AI, Multimodal Models
Tooling: HuggingFace, Weights & Biases, Git, Linux